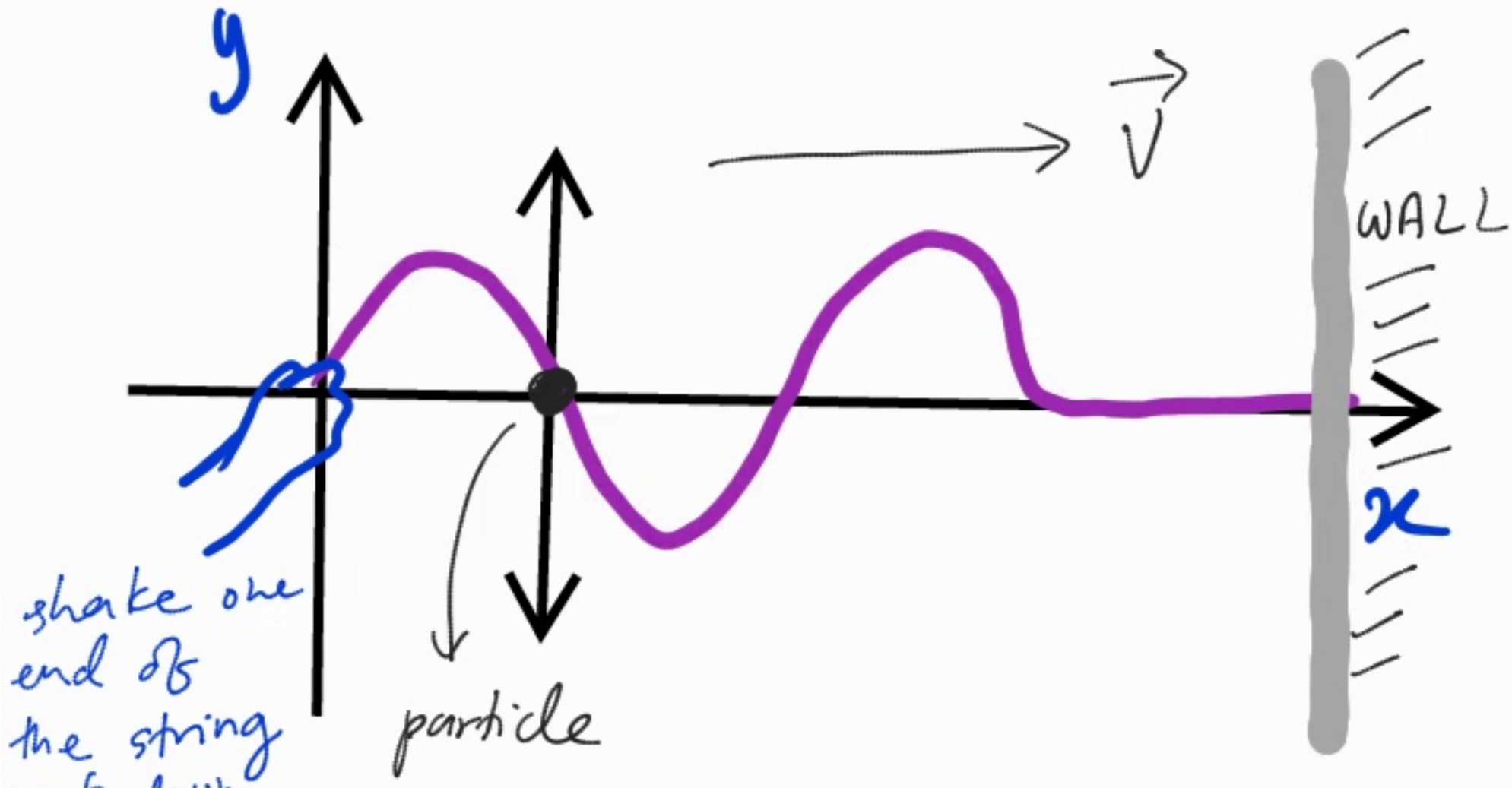
Transverse Waves

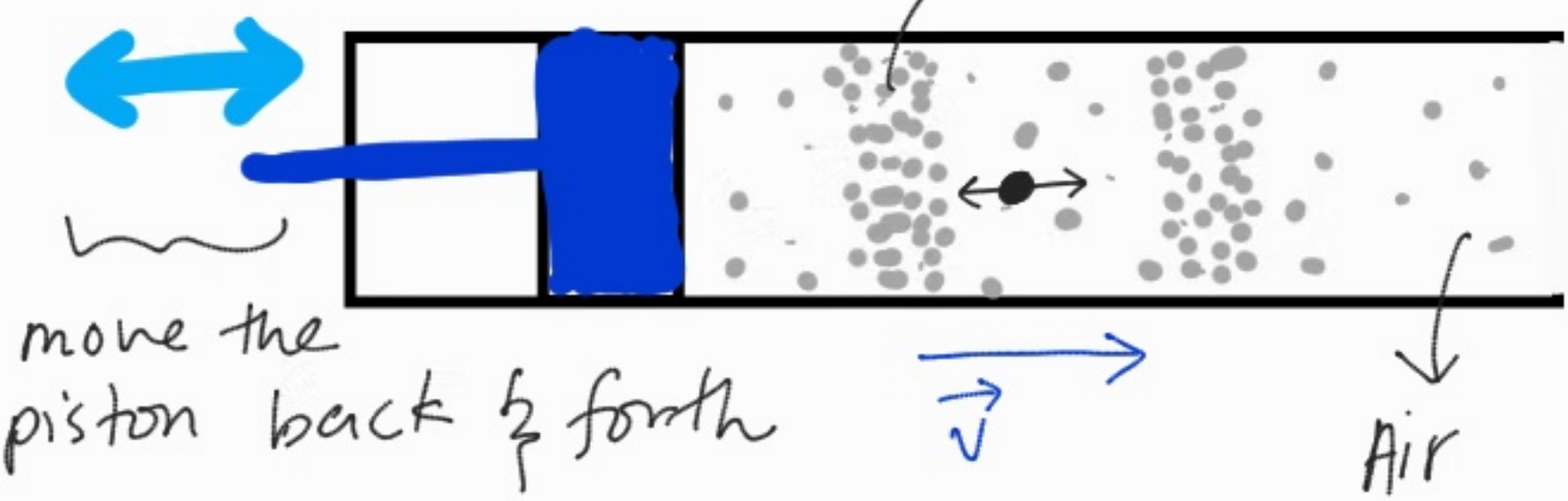
- Consider a string example:



- In this example, each particle of the string moves up & down in a transferable way (to the motion of the wave).
- Also known as travelling waves.

Longitudinal Wave.

- Let's think about another scenario where we generate waves using a piston-cylinder device. Compression of air particles



- Each particle in the fluid moves back & forth parallel to the motion of the wave.
- fluid particles oscillate horizontally.

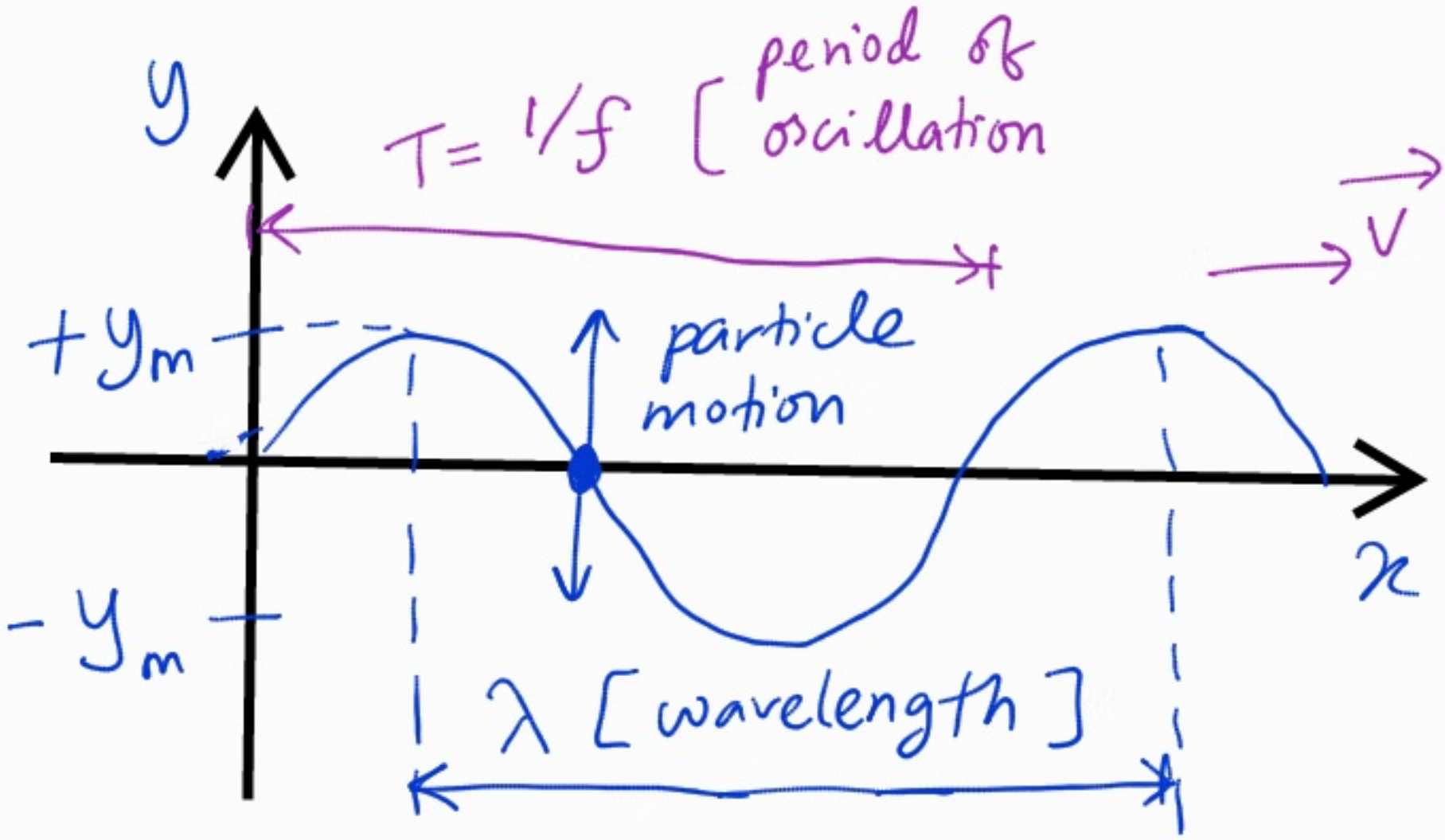
Harmonic Wave Function

⇒ **Definition:** a function that completely describes a travelling wave on a string and the motion of any element along its length.

$$y(x,t) = y \sin(kx - \omega t + \phi)$$

- $y(x,t)$: displacement
- y_m: Amplitude
- k: angular wave number
- x: position
- ω: angular frequency
- t: time
- ϕ: phase angle

Harmonic Wave Function



- String motion example: where the wave moves at constant speed along the length of the string.
- The particle motion in the string experiences simple harmonic motion perpendicular to the motion of the wave, $\vec{v}$.

Harmonic Wave Function

$$y(x,t) = y_m \sin(kx - \omega t + \phi)$$

- The angular wave number, k is related to the particle's motion with respect to its wave length, λ:

$$k = \frac{2\pi}{\lambda}$$

→ analogous to: $\omega = \frac{2\pi}{T} = 2\pi f$

Note: k is not a constant. It depends on the wave characteristics.

Harmonic Wave Function

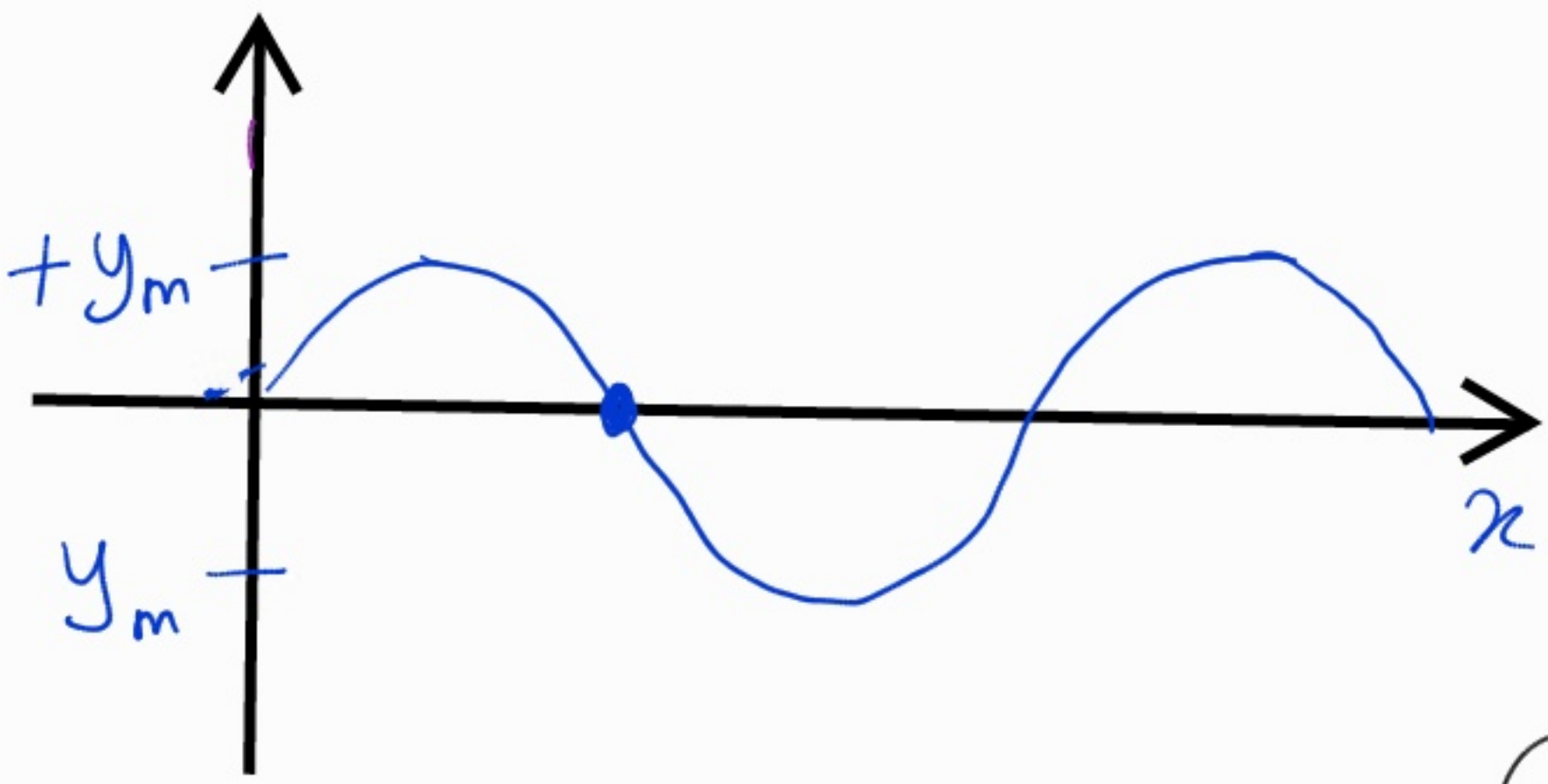
- Let's analyze the wave when the phase constant is varied.

I : Set $\phi = 0$

At $t = 0$, $y = y_m \sin(kx + 0)$

$= y_m \sin(kx)$

$\therefore$ at $x = 0$, $y = 0$

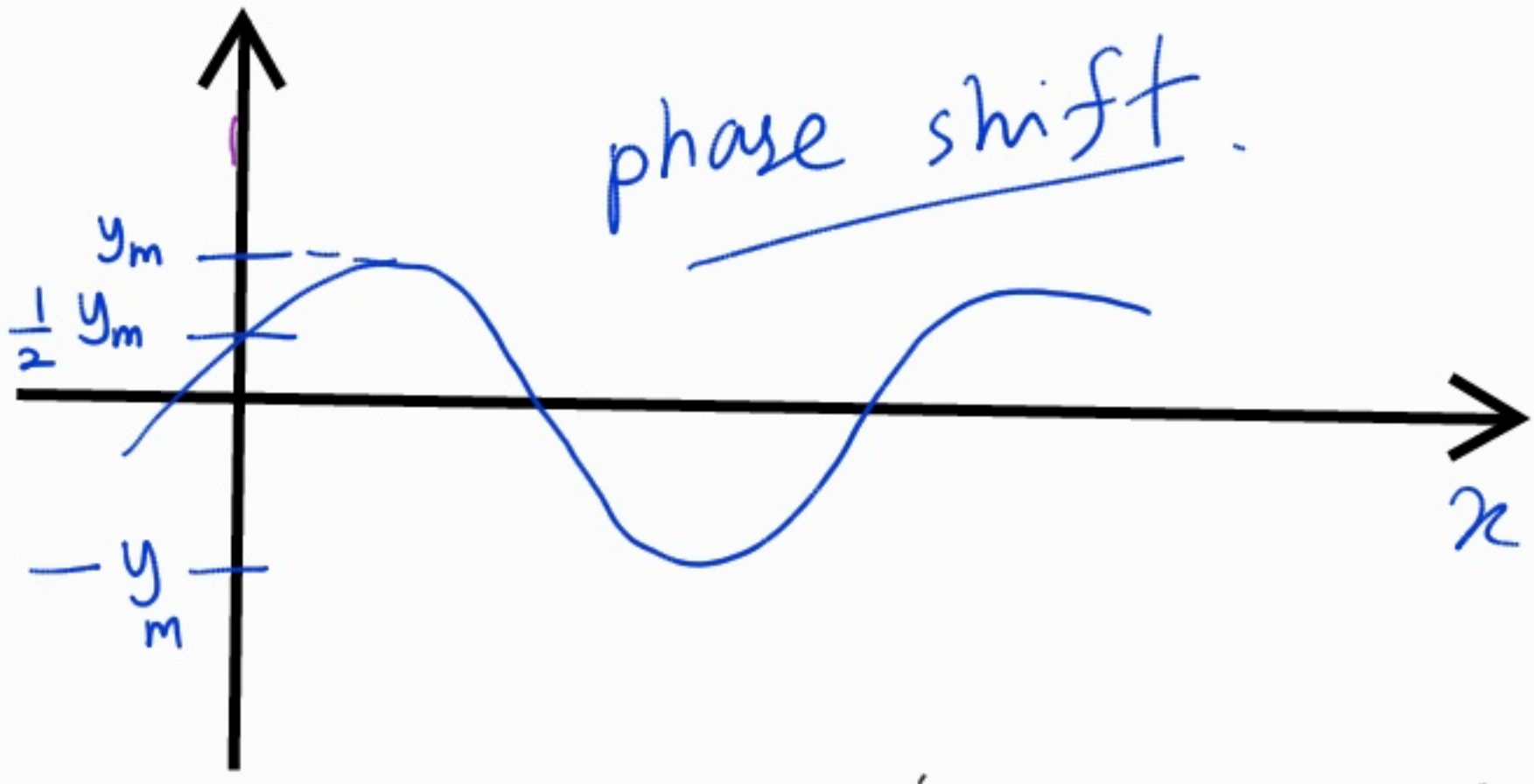


Harmonic Wave Function

- How about when $\phi = \pi/6$? Then

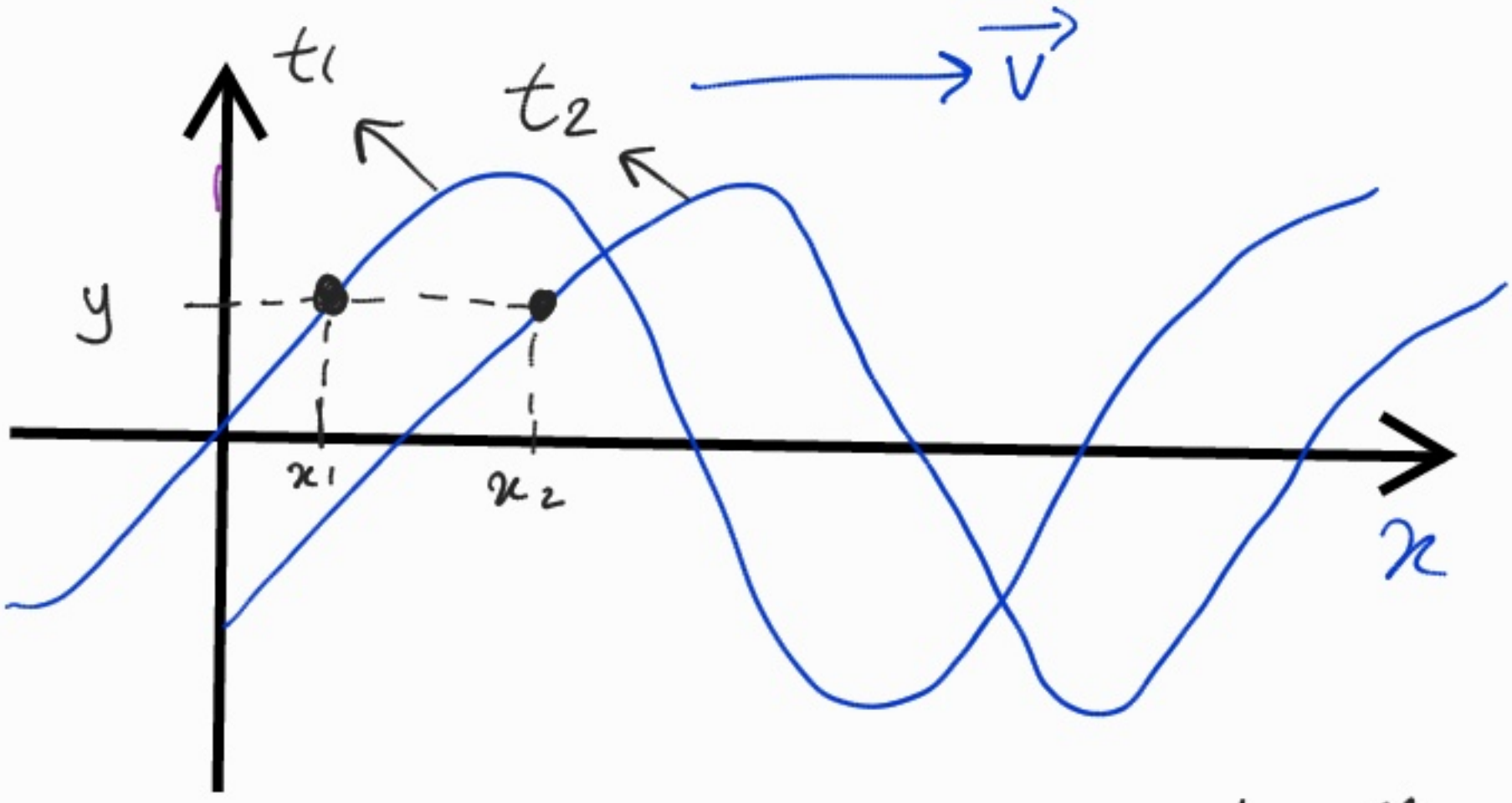
at $t=0$, $y = y_m \sin(kx + \pi/6)$

At $x = 0$; $y = y_m \sin\left(\frac{\pi}{6}\right) = \frac{y_m}{2}$



- Let's determine the expression for the speed of this travelling wave.

Harmonic Wave Function



- Frame-by-frame view of a travelling wave.
- Notice at x_1 & x_2 the displacement(y) is the same. $y = y_1 = y_2$
- Let's get the expression for its travelling velocity.

NEXT PAGE

Harmonic Wave Function

Since $y_1 = y_2$ and let $\phi = 0$;

then:

$$y_m \sin(kx_1 - \omega t_1) = y_m \sin(kx_2 - \omega t_2)$$

$$kx_1 - \omega t_1 = kx_2 - \omega t_2$$

$$kx_1 - kx_2 = \omega t_1 - \omega t_2$$

$$k\Delta x = \omega \Delta t$$

$$\therefore \frac{\Delta x}{\Delta t} = \frac{\omega}{k} = v$$

- Another way to express this:

$$v = \frac{\omega}{k} = \frac{2\pi/T}{2\pi/\lambda} = \frac{\lambda}{T}, \text{ since } T = \frac{1}{f}$$

$$\boxed{v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f}$$

Harmonic Wave Function

- This means for a travelling wave

$$y(x,t) = \underset{m}{y} \sin(kx \pm \omega t + \phi) :$$

(-) → wave travelling in the $+x$ direction.

(+) → wave travelling in the $-x$ direction.

- Let's derive the governing differential equation for the wave function :

$$y(x,t) = \underset{m}{y} \sin(kx - \omega t + \phi)$$

NEXT PAGE →

The Wave Equation.

$$y = y_m \sin(kx - \omega t + \phi)$$

⇒ Let's begin by finding its partial differentials with respect to space, x and time, t.

$$\frac{\partial y}{\partial x} = k y_m \cos(kx - \omega t + \phi)$$

$$\frac{\partial^2 y}{\partial x^2} = -k^2 \underbrace{y_m \sin(kx - \omega t + \phi)}_{y}$$

$$= -k^2 y$$

⇒ Repeat the procedure with t.

NEXT PAGE →

The Wave Equation

$$\frac{\partial y}{\partial t} = -\omega y_m \cos(kx - \omega t + \phi)$$

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2 \overbrace{y_m \sin(kx - \omega t + \phi)}^{y}$$

$$= -\omega^2 y$$

For x:

$$y = \frac{-1}{k^2} \frac{\partial^2 y}{\partial x^2}$$

For t:

$$y = \frac{-1}{\omega^2} \frac{\partial^2 y}{\partial t^2}$$

Therefore:

$$\frac{\partial^2 y}{\partial t^2} = \frac{\omega^2}{k^2} \cdot \frac{\partial^2 y}{\partial x^2}$$

The Wave Equation.

- Since the velocity of a travelling wave is: $v = \frac{\omega}{k}$,

Therefore;

$$\frac{\partial^2 y}{\partial t^2} = \frac{\omega^2}{k^2}\frac{\partial^2 y}{\partial x^2} = v^2\frac{\partial^2 y}{\partial x^2}$$

$$\boxed{\frac{\partial^2 y}{\partial t^2} = v^2\frac{\partial^2 y}{\partial x^2}}$$

THE WAVE EQUATION.

Practice Problem [1]

Radio waves transmitted by a spacecraft through space have a wavelength, $\lambda = 0.15\,m$. What is the frequency of the transmission?

Hint: Radio waves and all electromagnetic waves travel at the speed of light.

Solution:

Radio waves travel at the speed of light: $V = C = 3 \times 10^8$ m/s

$\lambda = 0.15$ m

Since $V = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f$

$\therefore \quad c = \lambda f \rightarrow f = \frac{c}{\lambda}$

$$f = \frac{3 \times 10^8 \text{ m/s}}{0.15 \text{ m}} = \underline{2000 \text{ MHz}}$$

Practice Probem 2

Consider the following waves:

$$y_1(x,t) = 2 \sin(4x - 2t)$$

$$y_2(x,t) = \sin(3x - 4t)$$

$$y_3(x,t) = 2 \sin(3x - 3t)$$

a) Rank the waves based on their wave speeds. (u)

b) Rank the waves based on their maximum speeds perpendicular to the wave's direction of motion. (v)

Solution.

a) $u = \frac{w}{k}$

$\therefore u_1 = \frac{2}{4} = 0.5 \quad ; u_2 = \frac{4}{3} = 1.333$

$u_3 = \frac{3}{3} = 1$

$\therefore \quad u_1 < u_3 < u_2$

b) $v = \frac{dy}{dt}$

$v_1 = 2 \frac{d}{dt} [\sin (4x - 2t)]$

$= 2 (-2) \cos (4x - 2t)$

$= -4 \cos (4x - 2t)$

Solution (Con't)

$$v_2 = \frac{dy_2}{dt} = \frac{d}{dt}\left[\sin(3x - 4t)\right]$$

$$= -4\cos(3x - 4t)$$

$$v_3 = \frac{dy_3}{dt} = \frac{d}{dt}\left[2\sin(3x - 3t)\right]$$

$$= -6\cos(3x - 3t)$$

$\therefore$ velocity amplitudes are:

$$|v_{1,m}| = 4 \; ; \; |v_{2,m}| = 4 \; ;$$

$$|v_{3,m}| = 6$$

$$\therefore \; v_{1,m} = v_{2,m} < v_{3,m}$$